Overview of ART Physics & Technologies

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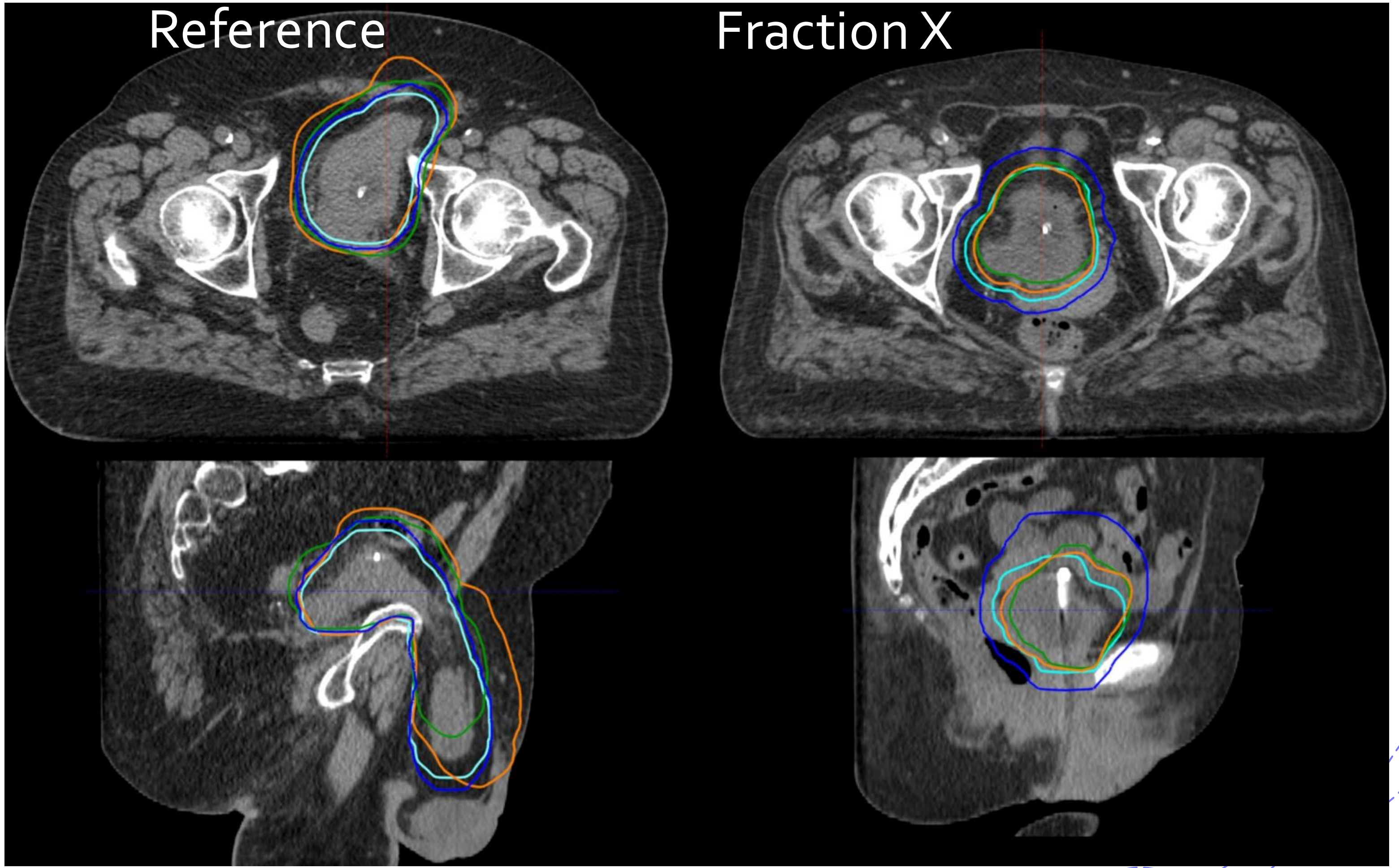
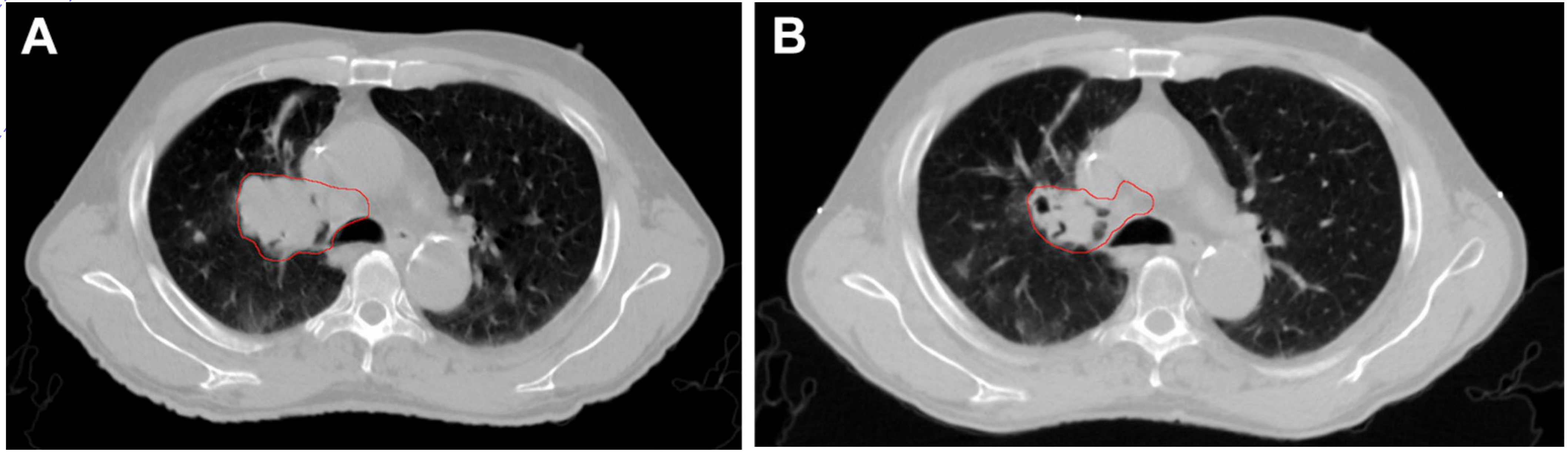
Known fact: Anatomy changes during RT



Online adaptive radiotherapy of urinary bladder cancer with full re-optimization to the anatomy of the day: Initial experience and dosimetric benefits, Åström et al, Radiotherapy and Oncology, 2022

Anatomy changes during RT



Ramella et al., Local control and toxicity of adaptive radiotherapy using weekly CT imaging: results from the LARTIA trial in stage III NSCLC - Journal of Thoracic Oncology, 2017

Traditional strategy in RT

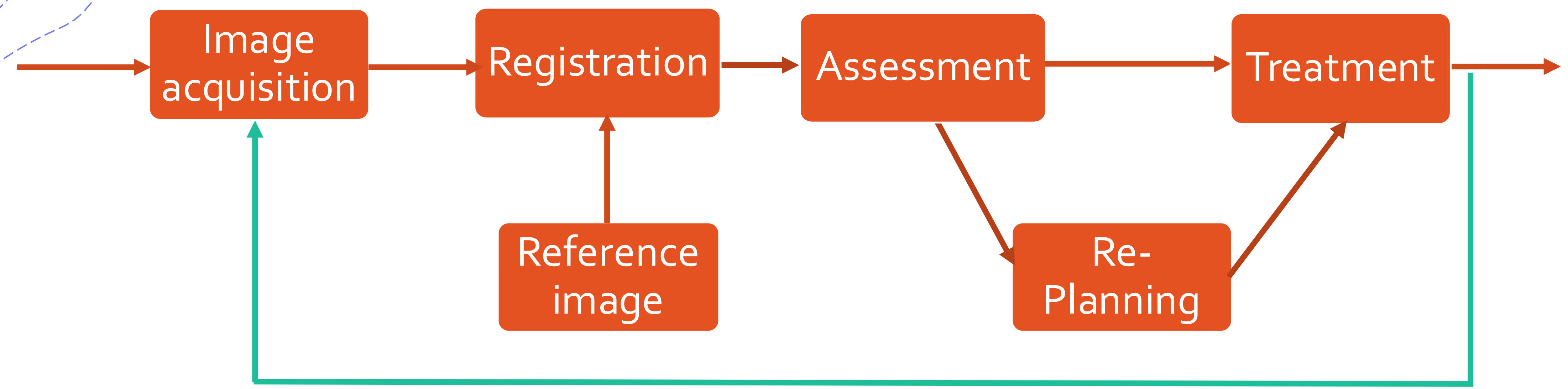
- And yet for decades, our default strategy has been: "plan once… and hope margins save us."
  - Increased toxicity
  - And/or Lower dose to the target

Adaptive Radiotherapy (ART)

First mentioned by Di Yan et al. in 1997

"Adaptive radiation therapy is a closed-loop radiation treatment process where the treatment plan can be modified using a systematic feedback of measurements. Adaptive radiation therapy intends to improve radiation treatment by systematically monitoring treatment variations and incorporating them to re-optimize the treatment plan early on during the course of treatment. In this process, field margin and treatment dose can be routinely customized to each individual patient to achieve a safe dose escalation"

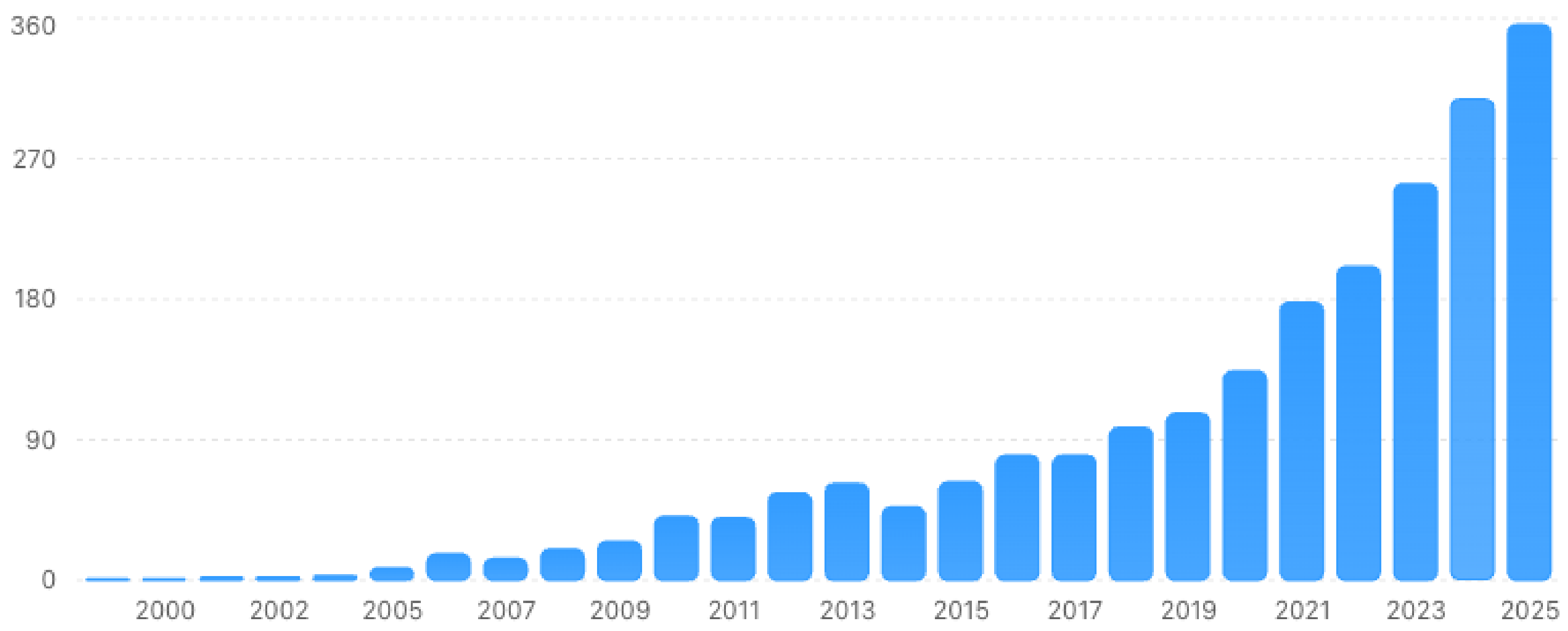
ART Daily Workflow



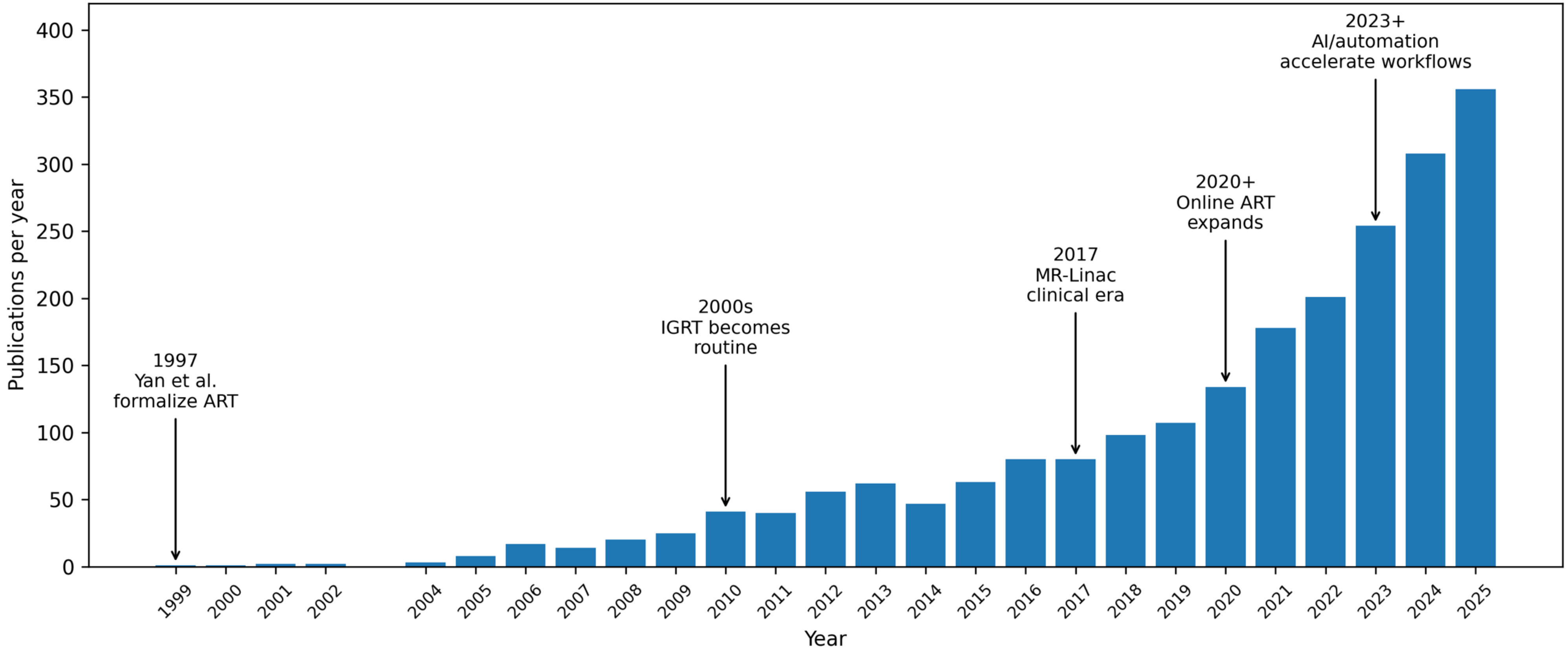
What best describes your institution's current experience with adaptive radiotherapy (ART)?

Growth of adaptive radiotherapy publications

PubMed publications containing the phrase 'adaptive radiotherapy' by year.



Growth of adaptive radiotherapy publications



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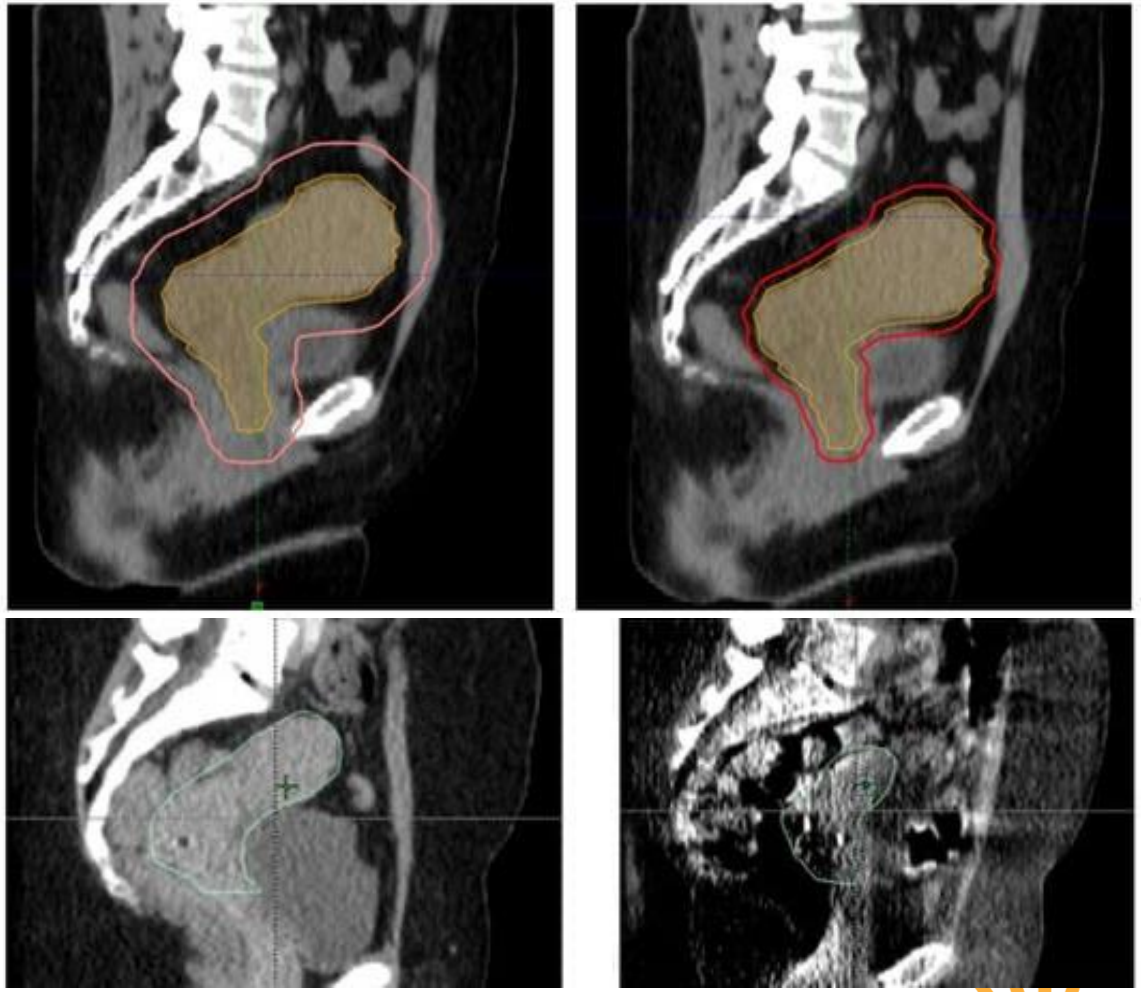
Potential benefits of ART



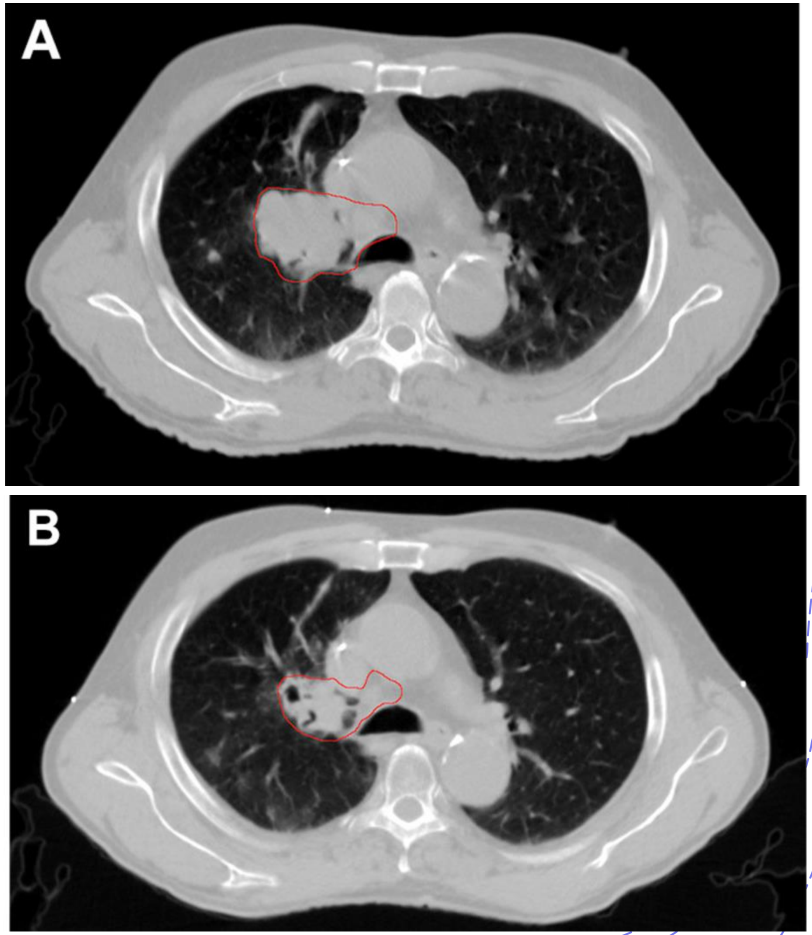
1. Very mobile anatomy in the pelvis
   - ART→ ability to reduce margins
2. Significant changes in size over the course of treatments
   - ART→ ability to account for these changes in tumor size

→ potentially treat less normal tissue while maintaining tumor coverage

Yen A, Shen C, Albuquerque K. The New Kid on the Block: Online Adaptive Radiotherapy in the Treatment of Gynecologic Cancers. Curr Oncol. 2023

Potential benefits of ART

- Reduced toxicity
- (Study found: 2% and 4% of acute and late ≥G3 lung damage vs RTOG 9410 concurrent arms (≥G3 toxicity rates of 13% and 17% in the standard and hyperfractionated arms, respectively))
  - Non-increased relapse compared to existing literature



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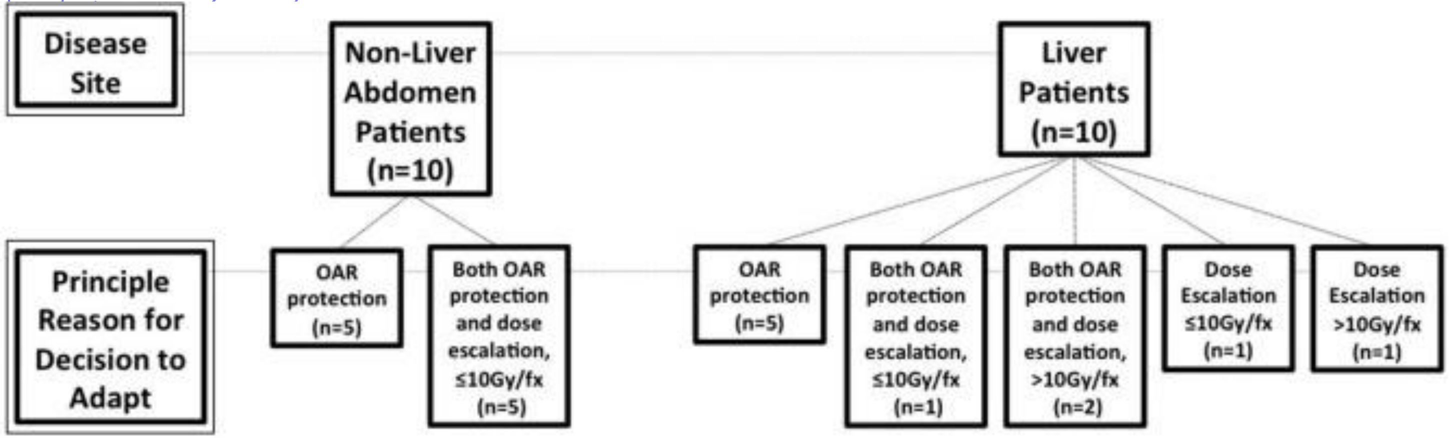
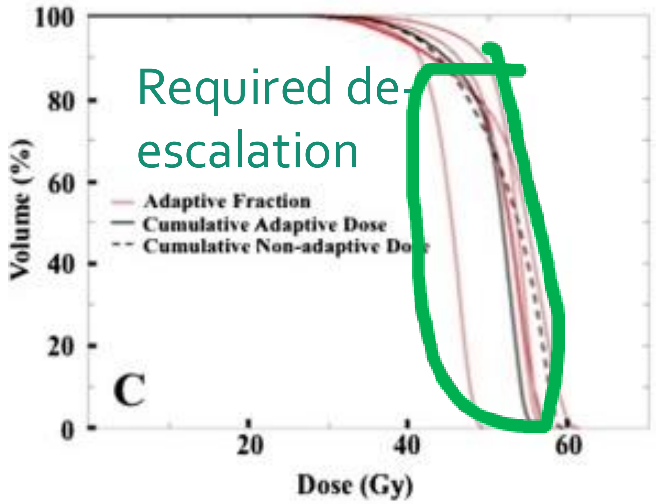
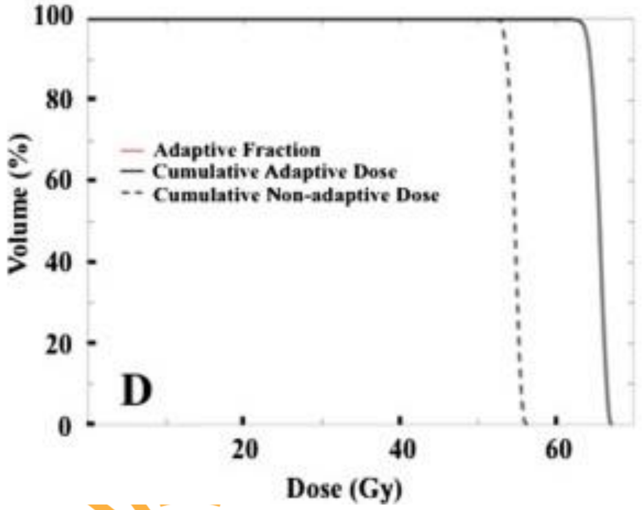
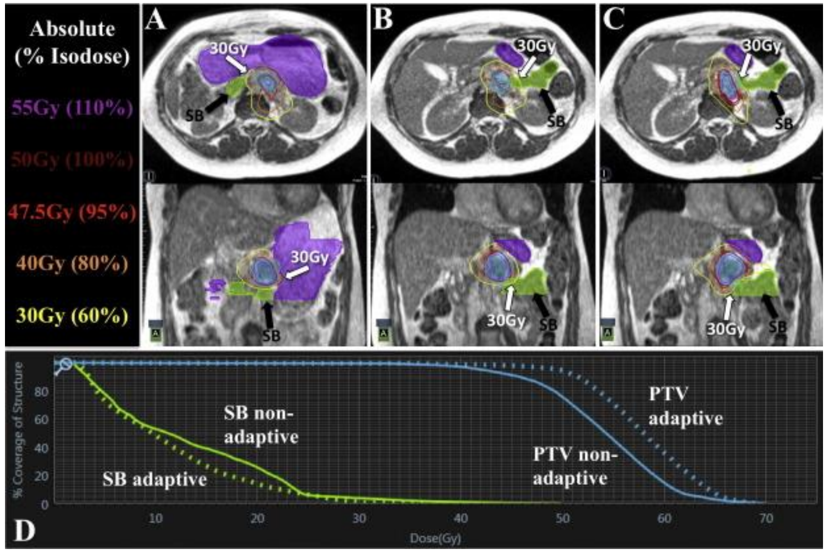
Potential benefits of ART







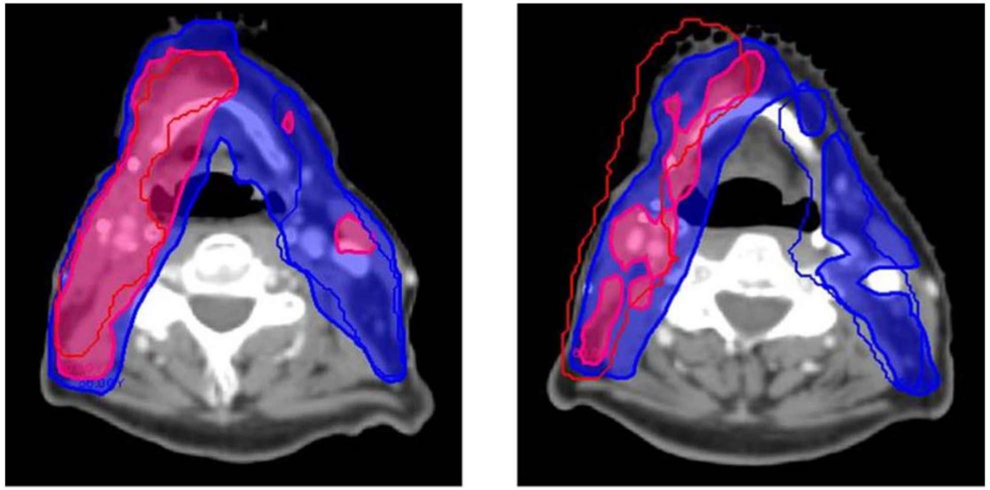


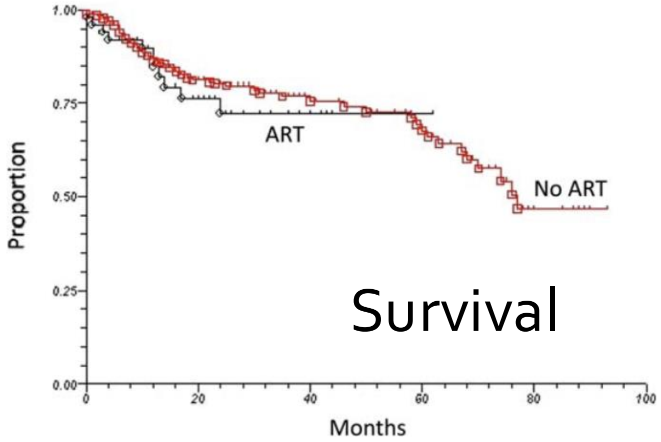
Henke et al, Phase I trial of stereotactic MR-guided online adaptive radiation therapy (SMART) for the treatment of oligometastatic or unresectable primary malignancies of the abdomen, Radiotherapy and Oncology 2018

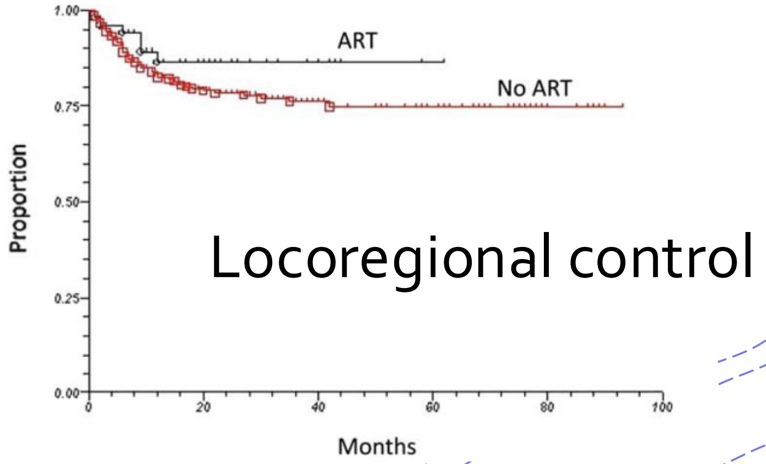
Potential benefits of ART

Initial Plan dose

Dose from original plan on anatomy of the day







Clinical outcomes among patients with head and neck cancer treated by intensity-modulated radiotherapy with and without adaptive replanning, Chen et al, Head Neck 2014

Types of Adaptive Radiotherapy

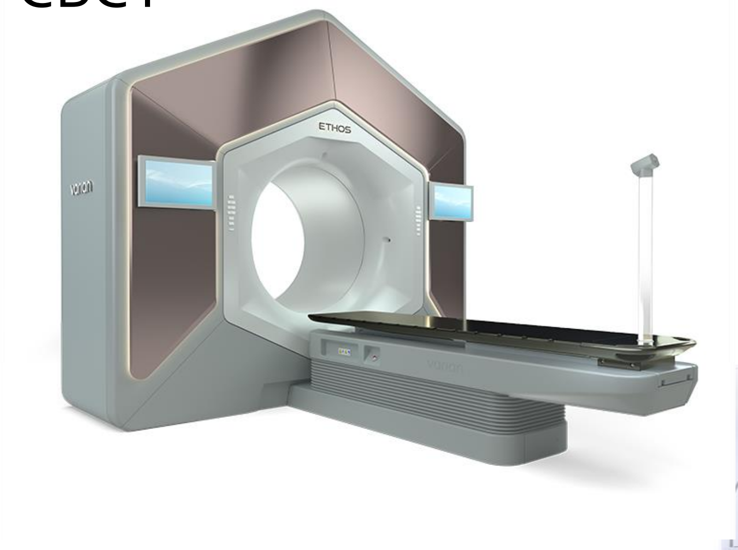
Offline ART	Online ART	Real-time ART
• Takes hours/days • Adaptation to slow changes • Infrequent (1x , 2x per treatment course) • Less resource intensive • Usually a combination of manual/automation work • Can use diagnostic images	• Takes minutes • Adaptation to daily changes • More frequent (daily, weekly) • Resource intensive • Needs higher degree of automation • Uses onboard images (can use other available additional images)	• Takes seconds • Adaptation to motion/intra-fraction changes • Requires real-time on-board images

Key elements for ART

- *Imaging*
- *Auto segmentation*
- *Evaluation and Decision-Making*
- *Dose Accumulation*
- *Adaptive planning*

Onboard IGRT systems

CBCT

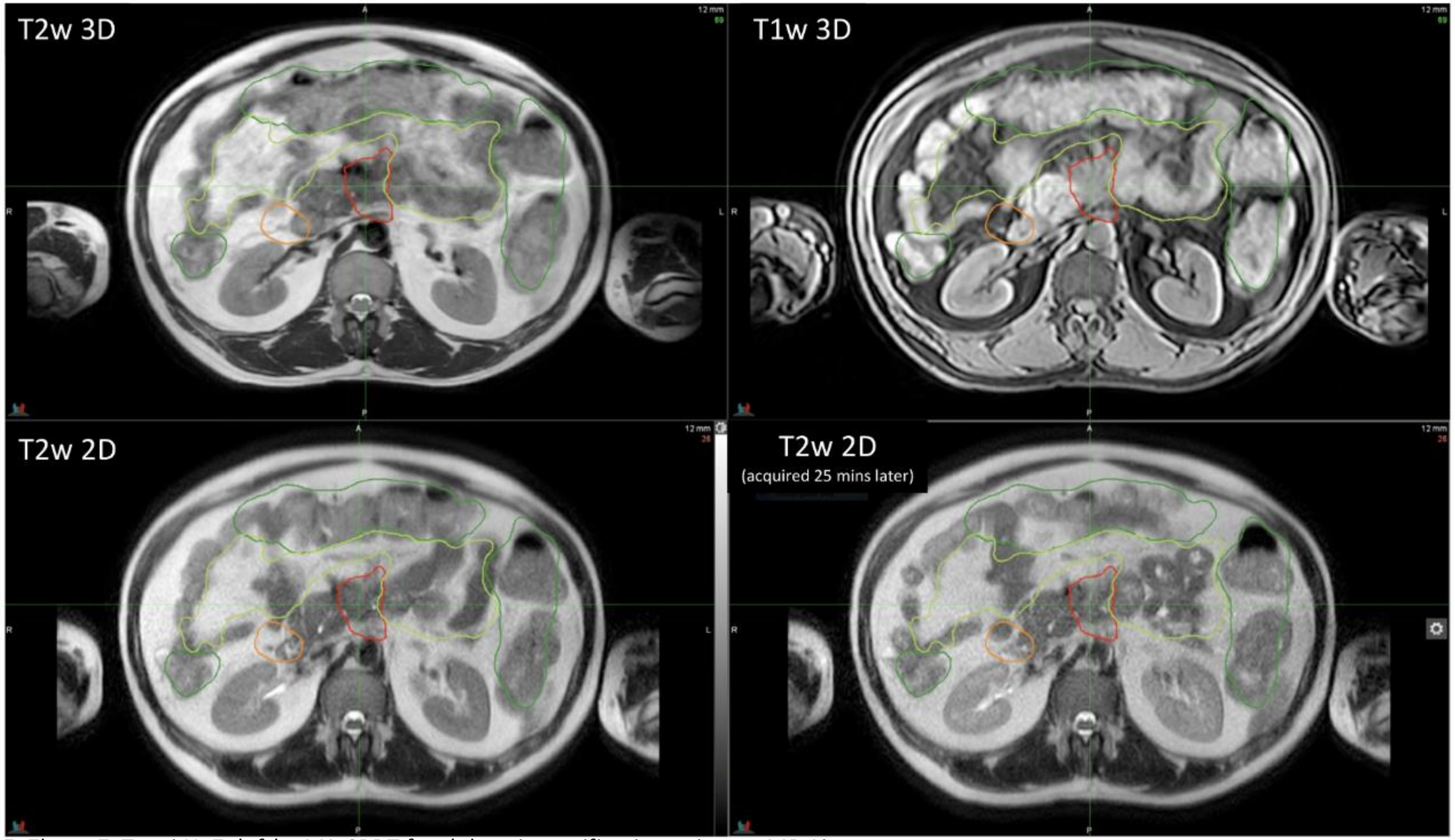


MR linac

PET linac

MR-based ART

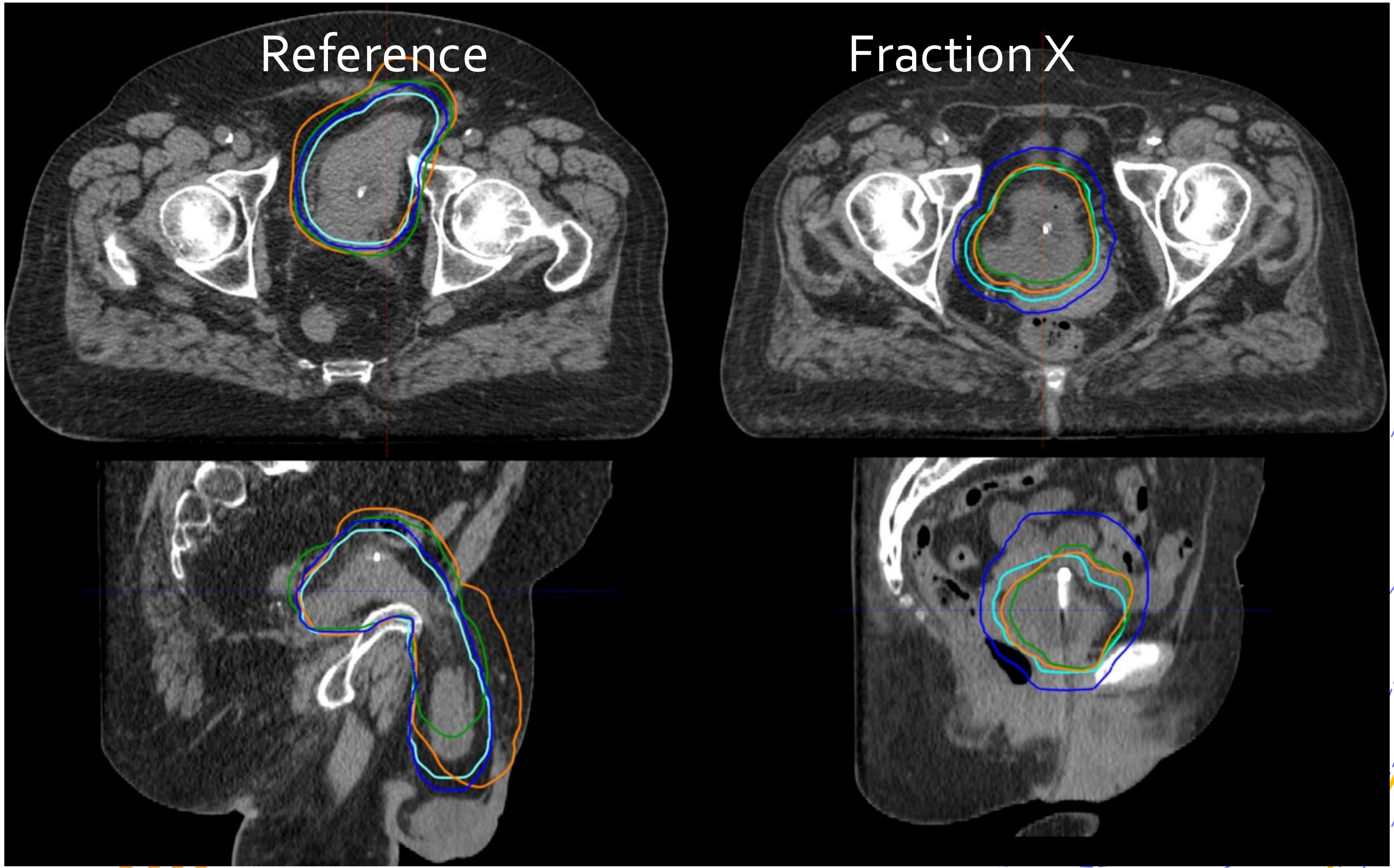
- Great soft-tissue contrast
- Non-ionizing radiation imaging
- Multiple MR sequences may be used for online contouring



Brennan VS, Burleson S, Kostrzewa C, Godoy Scripes P, Subashi E, Zhang Z, Tyagi N, Zelefsky MJ. SBRT focal dose intensification using an MR-Linac adaptive planning for intermediate-risk prostate cancer: An analysis of the dosimetric impact of intra-fractional organ changes. Radiother Oncol. 2023

kV-based ART

- kV more widely available
- Good fit for many applications
- Benefits from improvements in CBCT quality (HyperSight, for example)



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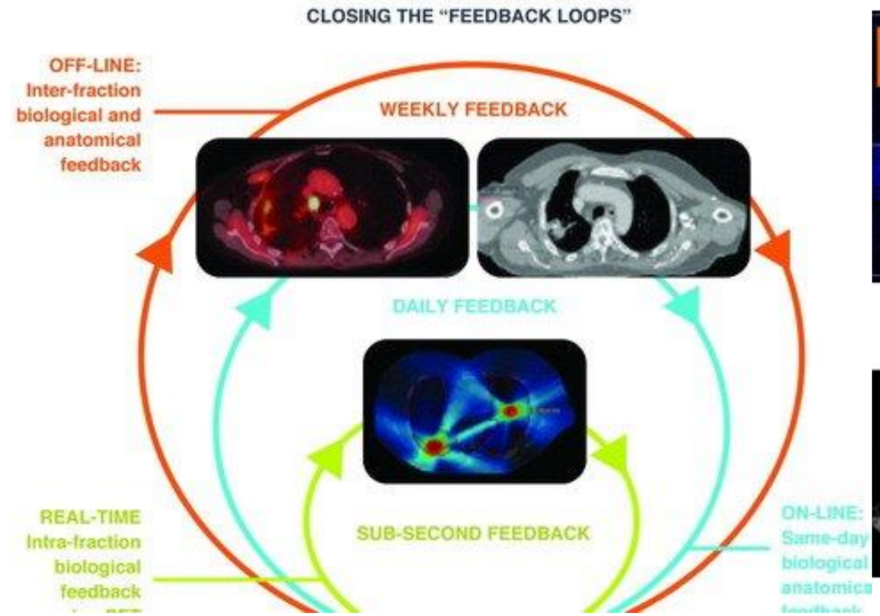
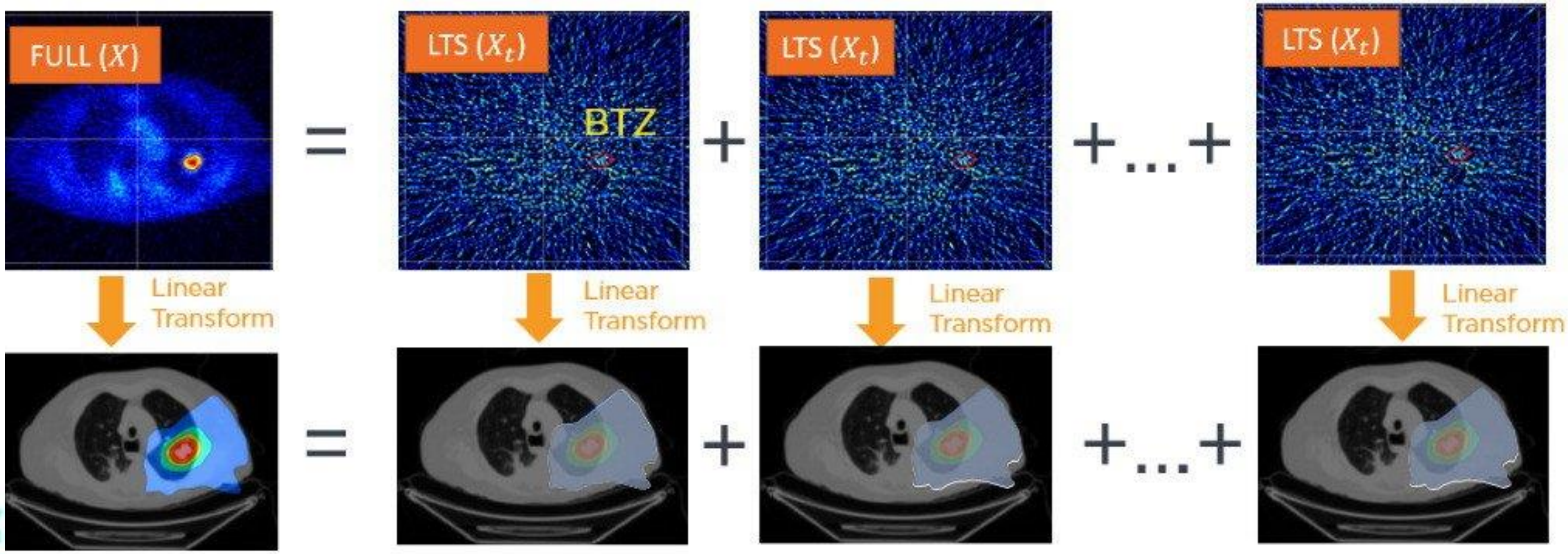
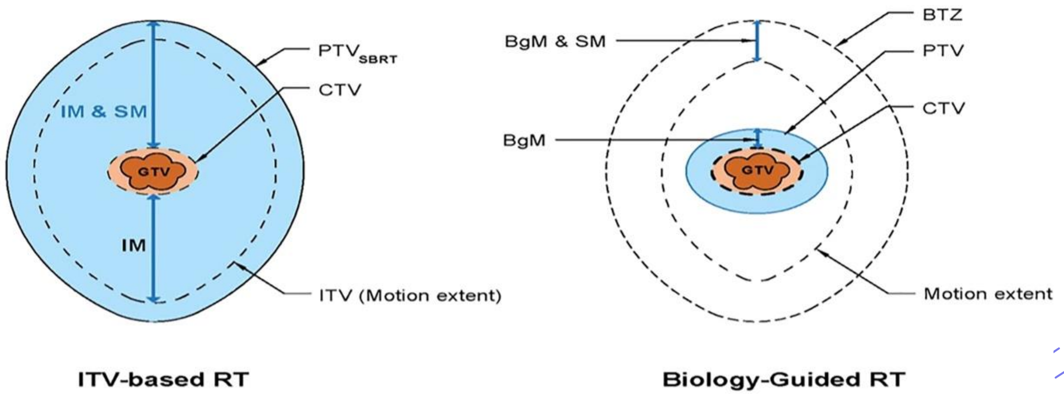
PET-based ART





Shirvani SM et al. Biology-guided radiotherapy: redefining the role of radiotherapy in metastatic cancer. Br J Radiol. 2021

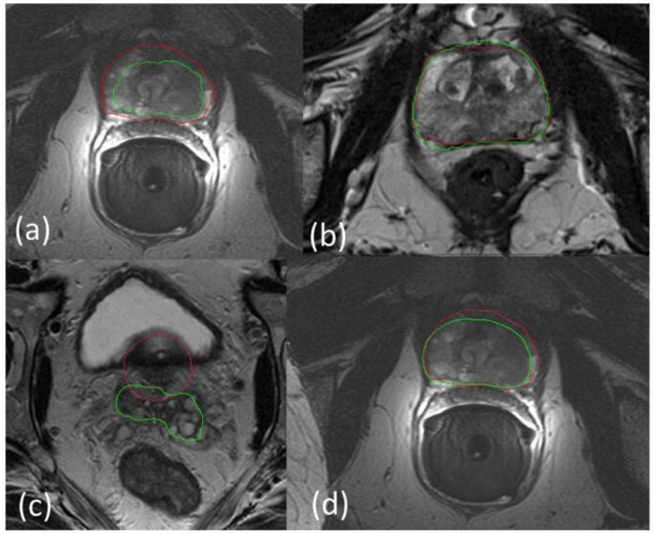


Seyedin et al. Frontiers, 2022

Segmentation process

Image Acquisition → Import → Registration → OAR Autoseg → Target volume → Planning

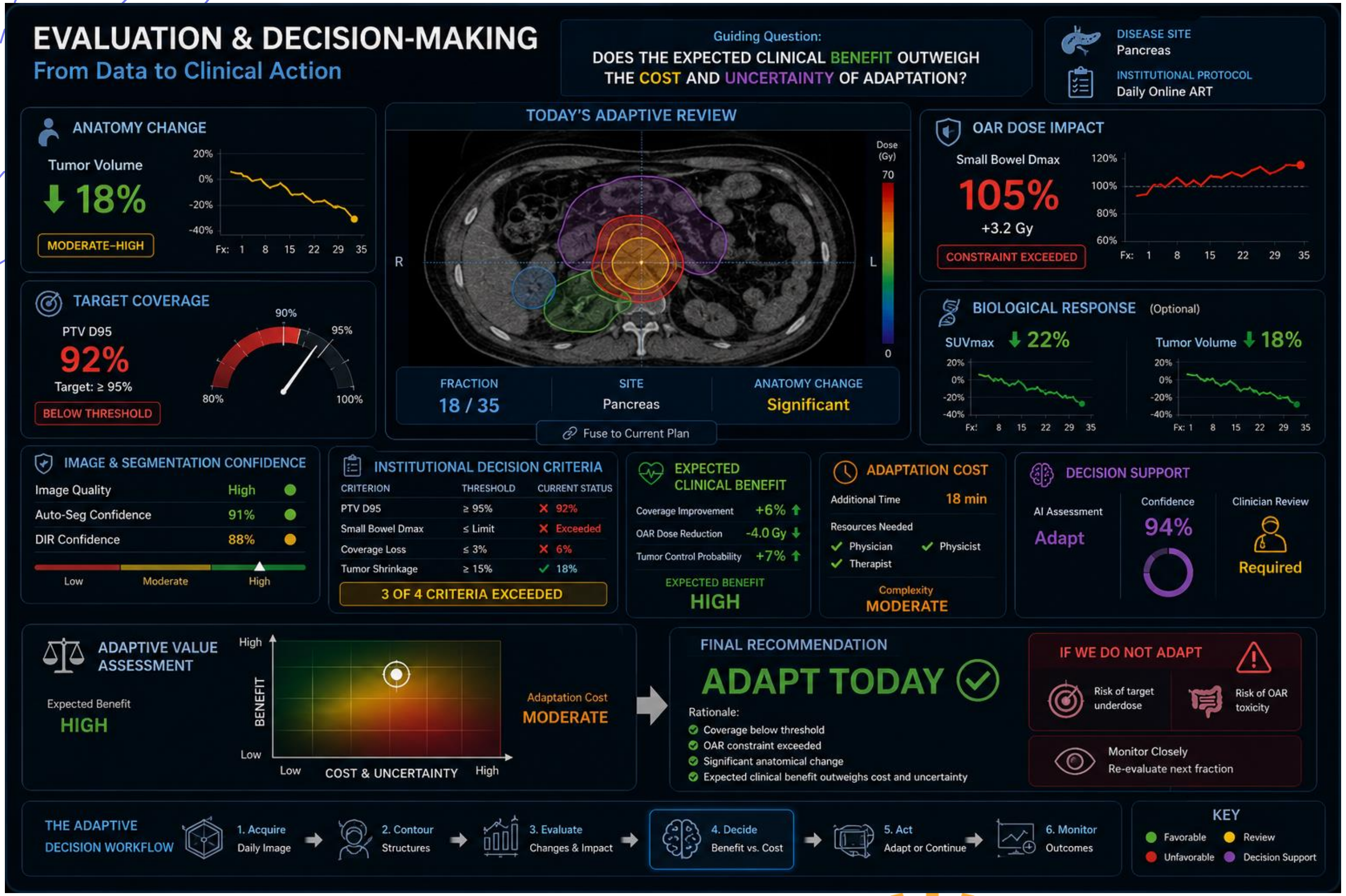
Deep learning-based auto-segmentation of targets and organs-at-risk for magnetic resonance imaging only planning of prostate radiotherapy, Elguindi et al, Phys Imaging Radiat Oncol, 2019



- Green – AI
- Red - clinical

- Most common: DIR (from prior image), atlas-based, and AI-based
- Reality: Commercial systems are catching up with developers for OAR segmentation
- Probable exception: target autosegmentation – most challenging part that still warrants research/development

Evaluation and Decision-Making

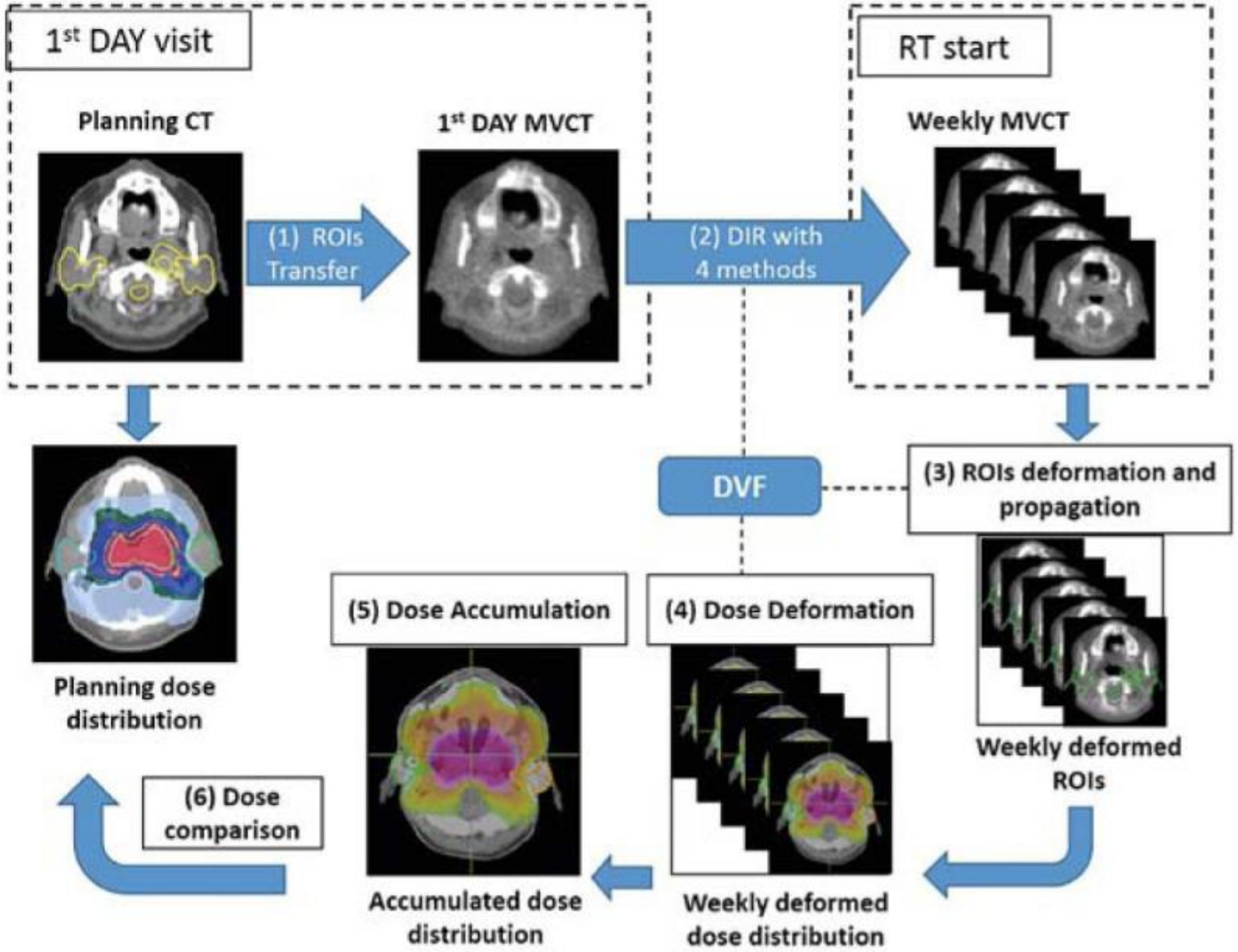


- Value-Based Adaptation

  Anatomical Change ≠ Adaptation

  Adapt when:
  ✓ Benefit is meaningful
  ✓ Risk is reduced
  ✓ Resources are justified

Dose Accumulation

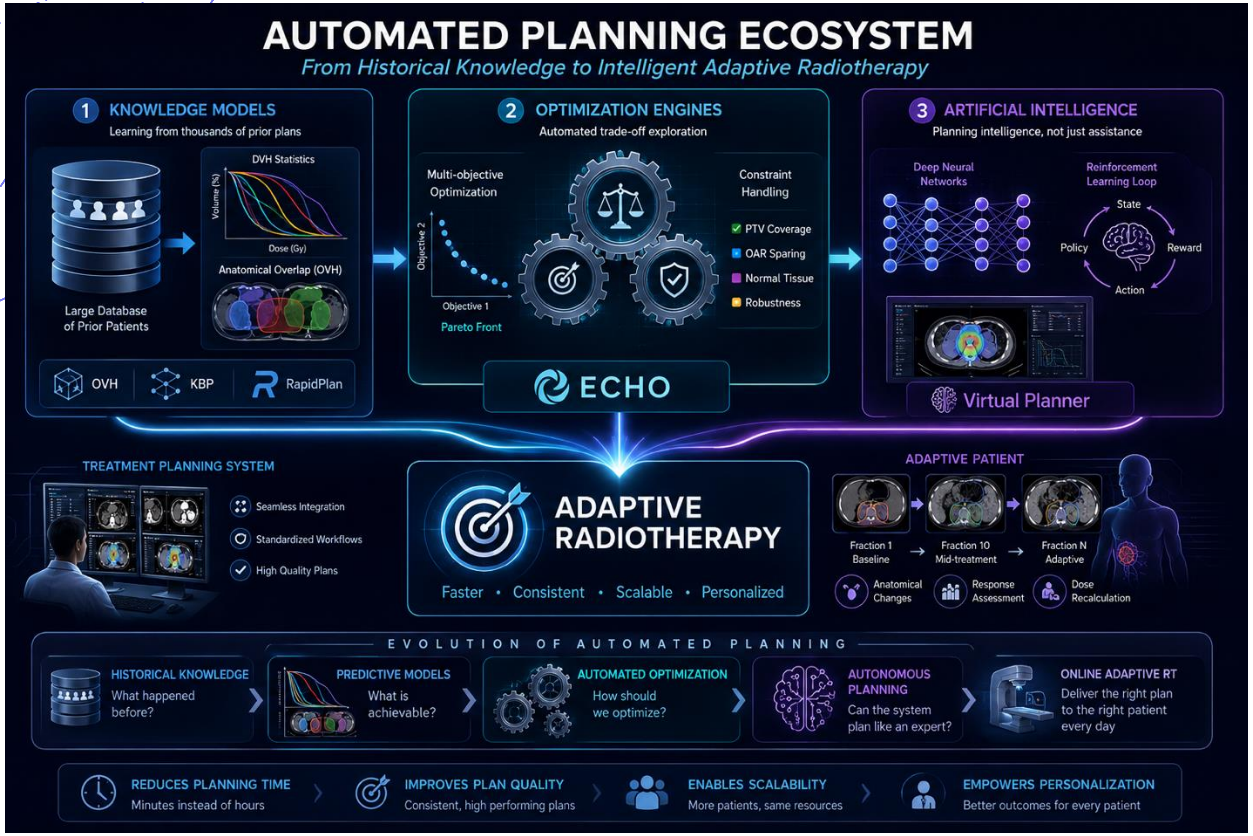


Areas where dose accumulation is important

- Assess delivered dose (vs planned dose)
- Adapt future doses based on what has been delivered so far
- Accuracy in TCP and NTCP models

Evaluation of Deformable Image Registration (DIR) Methods for Dose Accumulation in Nasopharyngeal Cancer Patients during Radiotherapy. Nobnop et al. Radiol Oncol. 2017

Planning automation



<u>Evolution of planning</u>

Historical Knowledge

↓

Prediction

↓

Hierarchical Optimization

↓

Intelligent Planning

Adaptive Radiotherapy

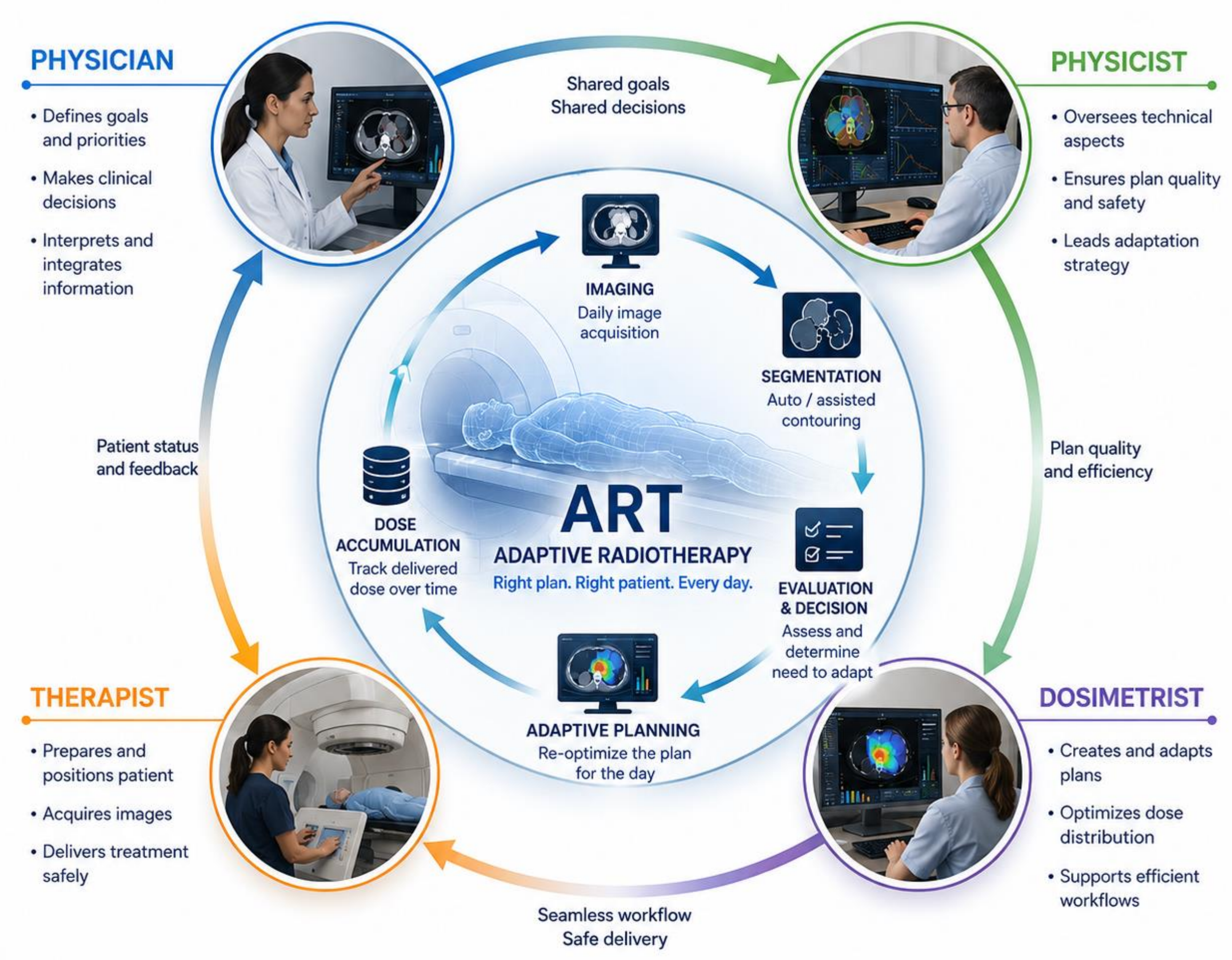
Wu, Binbin, et al. "Patient geometry-driven information retrieval for IMRT treatment plan quality control." *Medical physics* 36.12 (2009)

Fogliata et al. "On the pre-clinical validation of a commercial model-based optimisation engine: application to volumetric modulated arc therapy for patients with lung or prostate cancer." *Radiotherapy and Oncology* 113.3 (2014)

Zarepisheh et al. "Automated intensity modulated treatment planning: The expedited constrained hierarchical optimization (ECHO) system." *Medical physics* 46.7 (2019)

Shen et al., "Operating a treatment planning system using a deep-reinforcement learning-based virtual treatment planner for prostate cancer intensity-modulated radiation therapy treatment planning", MedPhys 2020

ART is a Team Sport

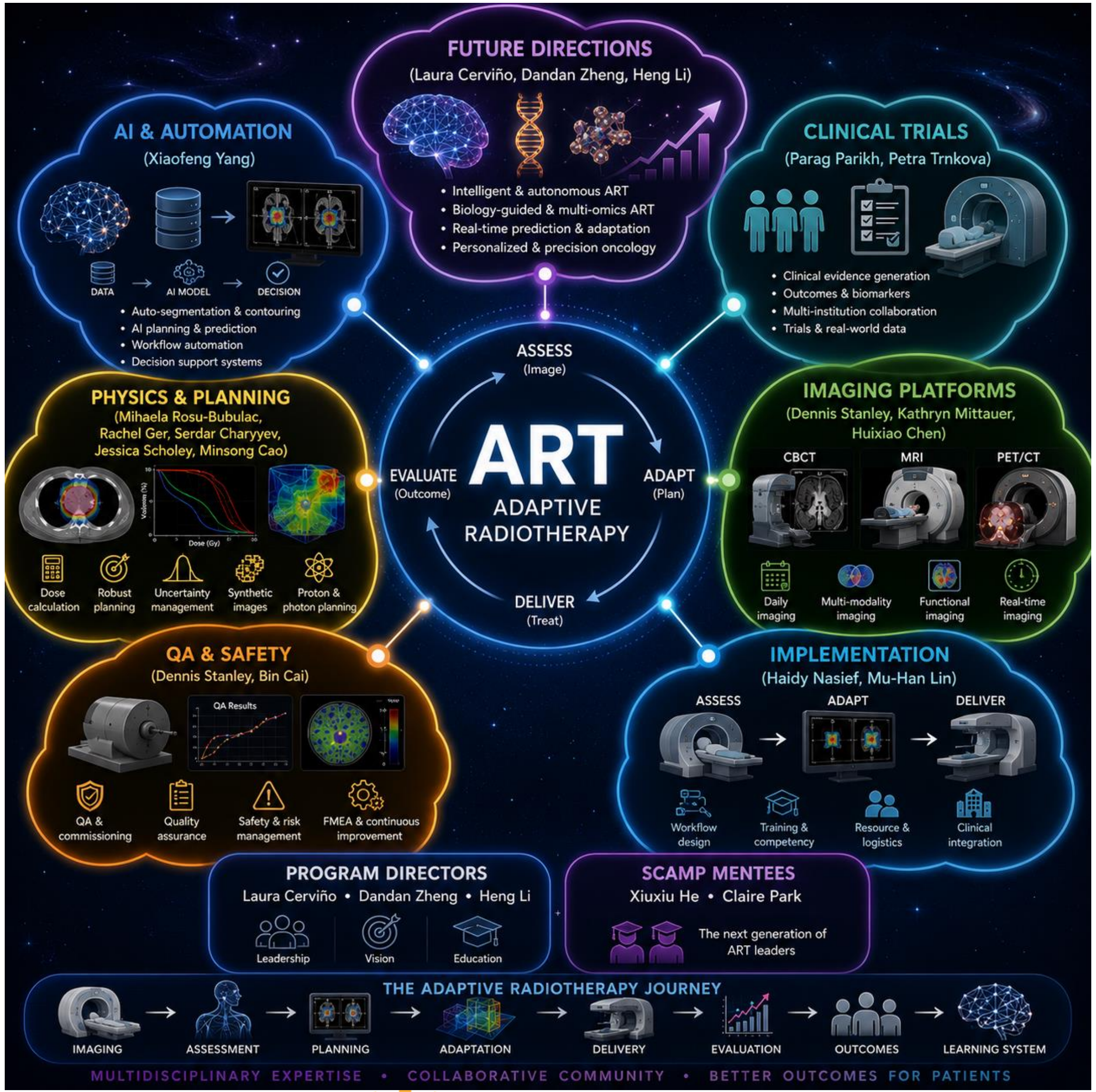


- ART is not a technology. ART is a workflow.

Challenges and considerations

- Logistical and operational demands
- Ethical and equity considerations
- Training and workforce development
- Infrastructure and implementation challenges
- Demonstrating clinical value

To end this introduction, what do you see as the biggest barrier to wider adoption of ART?



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